DeepMutate-3D: interactive protein language model mutation scanning on predicted structures

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Code: https://github.com/Saadman/DeepMutate-3D

Web application: https://huggingface.co/spaces/ras1992/DeepMutate-3D

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Abstract

Motivation. Protein language models predict the functional effect of amino acid substitutions without training on experimental data, but their output is a table of numbers. The structural context that makes such a table interpretable, which residues sit in an active site, a buried core, or a flexible loop, lives in separate software. Bridging the two currently requires local installation, model weights and scripting, which places the method out of reach of many laboratory scientists.

Results. DeepMutate-3D scores all 19 possible substitutions at every position of a protein using ESM-2 log-likelihood ratios, condenses them into a per-residue sensitivity score, retrieves the matching AlphaFold model, and paints the scores onto the structure as an interactive heat map in a web browser. Predictions are zero-shot. On the ProteinGym substitution benchmark (217 assays, 696,311 measured variants) it reaches a mean Spearman correlation of 0.425, within the range reported for ESM-2 650M. On 38,901 ClinVar variants across 2,011 proteins it separates pathogenic from benign substitutions with a mean per-protein AUROC of 0.881. Applied to proteins with independent ground truth, it places the six TP53 cancer hotspots among the most constrained positions of the DNA-binding domain (permutation $p = 0.009$) and ranks all eight cysteines of lysozyme C's four disulfide bridges within the ten most constrained of 147 positions (hypergeometric $p = 1.0e-11$).

Two comparisons made during validation generalise beyond this tool. Increasing model capacity buys more than refining the scoring protocol: ESM-2 650M under single-pass scoring exceeds ESM-2 150M under per-residue masked scoring while using two orders of magnitude less computation. And benchmark rank correlation does not predict functional-site recovery: two scoring modes separated by 0.007 mean Spearman on ProteinGym differ markedly on the case studies, recovering eight versus six of the lysozyme disulfide cysteines.

Availability. https://huggingface.co/spaces/ras1992/DeepMutate-3D runs in a browser with no installation or account, on shared NVIDIA hardware. Source, validation scripts and all result files are at https://github.com/Saadman/DeepMutate-3D under Apache-2.0.

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1. Introduction

Determining which residues in a protein tolerate mutation is a recurring question in variant interpretation, enzyme engineering, and functional annotation. Deep mutational scanning answers it experimentally but requires months of work per protein. Computational predictors offer a fast alternative, and protein language models in particular have proven capable of zero-shot variant effect prediction: trained only to reconstruct masked residues across evolutionary sequence data, they implicitly learn the constraints that shape protein families [1, 2].

The practical bottleneck is not accuracy but access and interpretation. A predictor returns a score per substitution. Interpreting that score requires knowing where the residue sits in the fold. A position scoring as highly constrained is only actionable once one knows it lines a catalytic pocket rather than a disordered terminus. Structural prediction has become universally available [3, 4], yet in practice the score and the structure are produced by different tools and joined by hand.

DeepMutate-3D closes that gap in a single interface. It is not a new scoring method. The underlying computation is the masked-marginal log-likelihood ratio of Meier et al. [2] applied to ESM-2 [1]. The contribution is integration and accessibility: a scan that requires no installation, no model weights, and no scripting, returning a structure that can be rotated and interrogated residue by residue, together with exportable tables and coordinate files.

Several established predictors address variant effect prediction, including alignment-based and language-model approaches evaluated in ProteinGym [5], and web servers exist for structure-based stability prediction. DeepMutate-3D does not compete with these on accuracy, and no head-to-head comparison against them is attempted here. Its distinction is the combination of zero-shot scoring, automatic structural mapping, and browser-based access with no installation.

Because the tool wraps an established method, its value depends on the wrapper being faithful. This note therefore reports validation on two independent benchmarks, an ablation across model sizes, a comparison of scoring protocols, and three case studies against ground truth that was established independently of this work.

2. Implementation

2.1 Scoring

For a sequence of length L, position i, and candidate residue m, the model computes a log-likelihood ratio against the wild-type residue:

```
LLR(i, m) = log P(m | context) - log P(wt_i | context)
```

Values near zero indicate that the model finds the substitution as plausible as the wild-type residue; strongly negative values indicate a substitution the model regards as unlikely given the evolutionary context. Because the quantity is a ratio, it is insensitive to positions at which the model is uniformly uncertain.

The 19 non-wild-type ratios at each position are averaged into a single Residue Sensitivity Score:

```
S_i = (1/19) * sum over m != wt_i of LLR(i, m)
```

Two scoring protocols are provided. *Masked marginals* replaces position i with a mask token before each forward pass, following Meier et al. [2], and costs one forward pass per residue. *Wild-type marginals* uses a single forward pass over the intact sequence, reading the output distribution at every position. Three ESM-2 sizes are selectable (35M, 150M, 650M parameters).

2.2 Sequences longer than the model context

ESM-2 has a 1,024-token context, which after special tokens accommodates 1,022 residues. Longer proteins are covered with overlapping windows of 1,022 residues at a stride of 766. Log-probabilities are accumulated in sequence space and divided by a per-position visit count, so that every interior residue is scored in two different contexts and averaged. The overlap avoids the systematic error that arises when a residue falls at a hard window boundary with no downstream context.

2.3 Structure retrieval and mapping

The predicted structure is retrieved from the AlphaFold Protein Structure Database [4] by UniProt accession, queried through the prediction API so that the current model version is used rather than a hard-coded filename. Sensitivity scores are written into the B-factor column of the PDB record, replacing pLDDT, which allows 3Dmol.js [6] to colour the cartoon representation by a numeric property without server-side rendering. The colour scale is a red-white-blue gradient with symmetric limits, placing the neutral midpoint at exactly zero so that colour is comparable across proteins. Exported coordinate files carry REMARK records documenting that the B-factor column no longer contains pLDDT.

2.4 Interface and deployment

The application is a single-file Gradio program. Hovering any point on the structure reports the residue's sensitivity score, its categorical verdict, and its most damaging and most tolerated substitutions. Results are exportable as a CSV table and as a scored PDB file that opens directly in PyMOL or ChimeraX.

The deployed instance runs on Hugging Face Spaces using dynamically allocated NVIDIA hardware, with GPU time reserved per request in proportion to the size of the job. The same source runs unmodified on CUDA, Apple Silicon, or CPU.

3. Validation

All evaluations are zero-shot. No experimental or clinical measurement is shown to the model at any point.

3.1 Mutation effect prediction

ProteinGym [5] aggregates deep mutational scanning experiments into a common benchmark. Predictions were generated for every measured single substitution and correlated with experimental fitness per assay using Spearman's rho.

Table 1. ProteinGym substitution benchmark, 217 assays, 696,311 variants, wild-type marginal scoring.

Model	Mean rho	Median rho	rho > 0.3	rho > 0.5	Compute
ESM-2 35M	0.291	0.321	51%	19%	17 s
ESM-2 150M	0.383	0.430	70%	35%	19 s
ESM-2 650M	**0.425**	0.477	77%	44%	47 s

The deployed configuration (650M) reaches 0.425. Published values for ESM-2 650M on this benchmark are in the region of 0.41 to 0.42. The present figure is therefore in the expected range, but it should not be read as outperforming those reports: scoring protocol, assay filtering, and aggregation scheme all differ between published evaluations, and the comparison here is indicative rather than exact. The purpose of Table 1 is to establish that the wrapper does not degrade the underlying method, not to claim an improvement over it.

3.2 Scoring protocol

Masked and wild-type marginal scoring were compared pairwise on an identical subset of 197 assays at 150M parameters.

Table 2. Paired comparison of scoring protocols, 197 assays, ESM-2 150M.

Protocol	Mean rho	Median rho	Compute
Wild-type marginals	0.392	0.441	11 s

Protocol	Mean rho	Median rho	Compute
Masked marginals	0.399	0.453	4,903 s

Masked marginals is superior on 72.1% of assays, but the paired mean difference is +0.0069 (95% CI +0.0050 to +0.0089) for 434 times the compute. The theoretically preferable protocol therefore buys a real but very small improvement at large cost.

Read alongside Table 1, this yields the practically useful observation that compute is better spent on model capacity than on scoring protocol: ESM-2 650M under wild-type marginals (rho 0.425) exceeds ESM-2 150M under masked marginals (rho 0.399) while using two orders of magnitude less computation. This comparison was performed at 150M; the corresponding measurement at 650M was not made, and the claim is not generalised beyond the model size tested.

3.3 Clinical variant classification

An independent task on independent data. Using the ClinVar substitution set curated within ProteinGym, variants of each protein were ranked by predicted damage and scored by AUROC against their pathogenic or benign annotation.

Table 3. ClinVar classification, 2,011 proteins, 38,901 variants (22,669 pathogenic), ESM-2 650M.

Metric	Value
Mean per-protein AUROC	**0.881**
Median per-protein AUROC	0.931
Proteins with AUROC > 0.7	89.4%
Pooled AUROC, per-protein z-scored	0.771
Pooled AUROC, raw scores	0.868

Per-protein AUROC is the primary metric because it matches the clinical question: given the variants observed in one gene, which are most likely to be damaging. Raw scores are not comparable between proteins, so pooled analysis requires per-protein standardisation; the pooled figure is lower than the mean per-protein figure because pooling mixes proteins of differing discriminability. Section 6 discusses the limitations of ClinVar as a benchmark.

3.4 Runtime

Table 4. Inference time, ESM-2 650M, identical proteins. CPU and Apple Silicon measured on an Apple M3 (6 threads for CPU); NVIDIA figures from dynamically allocated Hugging Face hardware, taking the minimum of repeated calls to exclude allocation overhead.

Protein	Length	Protocol	M3 CPU	M3 GPU (mps)	NVIDIA
Haemoglobin alpha	142	wild-type	0.41 s	0.09 s	0.18 s
Haemoglobin alpha	142	masked	30.98 s	7.82 s	1.45 s

Protein	Length	Protocol	M3 CPU	M3 GPU (mps)	NVIDIA
p53	393	wild-type	0.78 s	0.18 s	0.18 s
p53	393	masked	236.19 s	62.27 s	10.99 s
SARS-CoV-2 spike	1,273	wild-type	3.50 s	0.87 s	0.31 s
SARS-CoV-2 spike	1,273	masked	not run	not run	152.33 s

Wild-type marginal scoring is fast on all hardware; the complete 1,273-residue spike protein is scored in 3.50 s on CPU and under a second on either GPU. GPU acceleration matters for masked marginal scoring, where it delivers a 5.4 to 5.7-fold speed-up at matched protein length and where the advantage grows with length because attention cost scales quadratically. On small jobs under the fast protocol, dynamically allocated GPUs are slower than a laptop, because a fixed allocation overhead of approximately one second dominates.

4. Worked examples

Benchmarks establish that the scores are calibrated. These three examples test whether they answer questions a researcher would pose, each against ground truth established independently of this work.

Both scoring modes are reported. This turns out to matter: Section 3.2 showed the two modes differ by only 0.007 mean Spearman on ProteinGym, yet they differ substantially here. Rank correlation across a whole deep mutational scan and recovery of a handful of functional residues are not the same task, and a benchmark that measures the first does not settle the second.

4.1 Prioritising positions for mutagenesis

Tumour sequencing has identified six positions accounting for a large share of TP53 missense mutations. Scanning the sequence alone, with no cancer data, places those six at a median constraint percentile of 92.5 under masked marginals and 93.4 under wildtype marginals, with four of six in the most constrained tenth of the protein under either mode.

Against 20,000 random six-position draws from the DNA-binding domain, where all six hotspots lie, $p = 0.009$ (masked). Drawing from the whole protein gives $p = 0.0007$, but that null is too permissive: TP53's disordered terminal regions are trivially unconstrained, so the domain-restricted value is the one to cite.

For a gene with an established hotspot catalogue the tool recovers what is already known. For one without, the same ranking provides a prioritised shortlist for panel design or functional follow-up.

4.2 Locating functional sites in uncharacterised proteins

Six enzymes were selected before scoring on two criteria: curated UniProt active site or binding site annotations, and a length within the model context. No enzyme was examined and then discarded.

Mode	Panel median percentile	Enzymes at $p < 0.01$
Masked marginals	86.8	4 of 6
Wildtype marginals	80.6	3 of 6

Under masked marginals, TEM-1 beta-lactamase reaches $p = 0.0001$, phosphoglycerate kinase 1 $p = 0.00005$, GAPDH $p = 0.00015$ and carbonic anhydrase 2 $p = 0.005$. Two enzymes fail in both modes. Lysozyme C has three annotated sites, too few for significance however well they rank. Cationic trypsin ($p = 0.081$ masked) has an annotation set mixing substrate-binding with catalytic residues, and binding residues sit under weaker evolutionary constraint than catalytic ones.

Carbonic anhydrase 2 illustrates the mode dependence: significant under masked marginals ($p = 0.005$) but not under wildtype ($p = 0.101$). Users pursuing functional site discovery should select masked marginals rather than accept the default.

4.3 Selecting positions for library design

Directed evolution requires the inverse question answered: which positions tolerate substitution. Lysozyme C, 147 residues, provides a check at the constrained end, because UniProt annotates four disulfide bridges (C24-C145, C48-C133, C82-C98, C94-C112) whose eight cysteines should be essentially unsubstitutable.

Mode	Disulfide cysteines in the ten most constrained positions	Exact p
Masked marginals	8 of 8	1.0e-11
Wildtype marginals	6 of 8	4.4e-07

Under masked marginals the remaining two positions are a buried glycine and a conserved tryptophan in the substrate-binding cleft. The model received sequence alone: no structure, no annotation, no experimental data.

One qualification applies. Cysteines are constrained as a class, reaching a median constraint percentile of 96.6 in lysozyme and 90.2 in trypsin against a non-cysteine median near 48, so part of this effect is generic rather than specific to disulfide bonding. A partial control exists in carbonic anhydrase 2, whose single cysteine forms no disulfide bond and reaches only the 40.5 percentile, below the protein median. This suggests the model distinguishes bonded from free cysteines rather than merely conserving the residue type, but with a single control it remains suggestive rather than established.

The tolerant end of the ranking carries no comparable external check. Positions reported as substitution-tolerant are the model's own output, and this work does not verify experimentally that they tolerate mutation. That claim awaits a prospective test.

5. Operation

A scan requires a UniProt accession. Leaving the sequence field empty retrieves the canonical sequence from UniProt [7] automatically, which guarantees that scores and structure share residue numbering; mismatched numbering is the most common source of silent error when the two are assembled by hand.

Three outputs are produced. The interactive structure provides spatial context, with a hover readout giving each position's score, verdict and extreme substitutions. The per-residue table, exportable as CSV, provides the ranking for downstream prioritisation. The scored PDB carries sensitivity in the B-factor column, so it opens directly in PyMOL or ChimeraX and can be coloured with a single command, which is the path from a browser scan to a publication figure.

No installation, model download or account is required. A 142-residue protein completes in approximately four seconds. Detailed usage documentation is maintained with the source code rather than reproduced here.

6. Limitations

The scoring method presented here is not novel. It is the established masked-marginal procedure of Meier et al. [2] applied to ESM-2, and Section 3.1 reproduces rather than improves upon published performance. The contribution is integration and accessibility, and the validation exists to demonstrate that the wrapper is faithful to the underlying method, not to advance it.

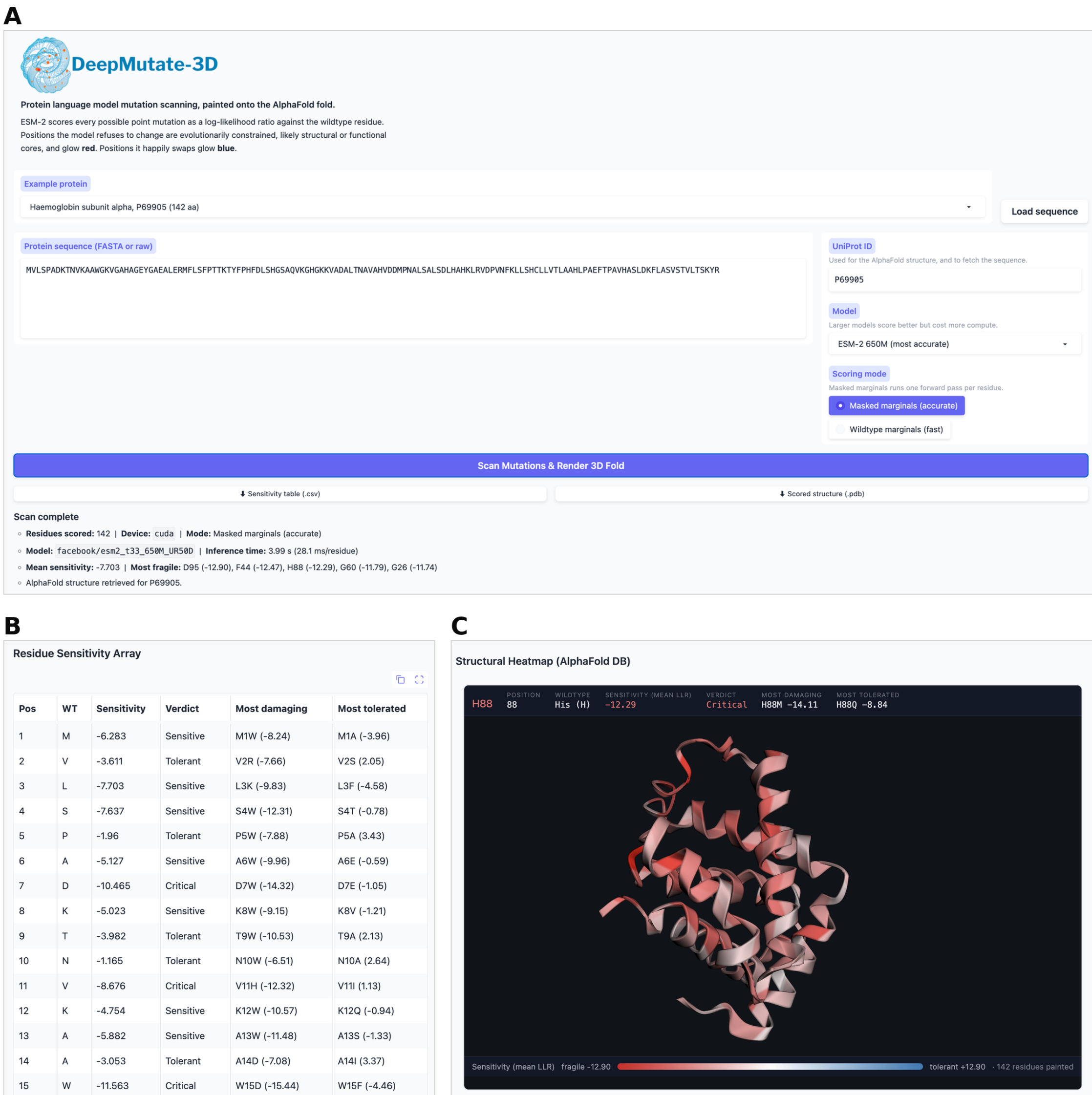
The benchmarks establish less than their headline figures suggest. ClinVar is an imperfect standard: pathogenic annotations are enriched in conserved functional domains and benign annotations in tolerant regions, which inflates apparent performance relative to prospective clinical use, and some submissions incorporate computational predictions, introducing partial circularity. The AUROC in Table 3 should therefore be read as evidence that the score behaves sensibly on clinically annotated variation, not as an estimate of diagnostic accuracy. Section 4 further shows that ProteinGym rank correlation does not predict functional-site recovery, since two scoring modes separated by 0.007 mean Spearman differ substantially on the worked examples. Any single benchmark number should be treated as narrower evidence than it appears. Within the worked examples, only the constrained end of the ranking is externally validated; positions reported as substitution-tolerant are the model's own output, and no experiment in this work confirms that they tolerate mutation.

Two classes of protein are poorly served. Designed sequences with no evolutionary history, such as engineered fluorescent proteins, score near zero at every model size, which is expected because the method estimates evolutionary constraint and there is none to estimate. Fast-evolving viral surface proteins also scored near zero at 150M parameters, but influenza haemagglutinin recovers to rho 0.46 at 650M, so that failure reflected model capacity rather than a property of the approach and may recede further with larger models.

Three narrower caveats apply to interpretation. Averaging the 19 substitutions at a position discards direction, since a residue may tolerate conservative changes while forbidding drastic ones, which is why the interface also reports the extreme substitutions at each position. AlphaFold monomer models are used throughout, so residues that matter only within a complex are under-weighted. For proteins beyond 1,022 residues, scores depend marginally on where sliding window boundaries fall.

Finally, and most importantly for anyone considering clinical application: this method predicts evolutionary constraint, which correlates with but is not identical to pathogenicity. DeepMutate-3D is a research tool for hypothesis generation and must not be used to inform diagnostic or treatment decisions.

7. Figures



Pos	WT	Sensitivity	Verdict	Most damaging	Most tolerated
1	M	-6.283	Sensitive	M1W (-8.24)	M1A (-3.96)
2	V	-3.611	Tolerant	V2R (-7.66)	V2S (2.05)
3	L	-7.703	Sensitive	L3K (-9.83)	L3F (-4.58)
4	S	-7.637	Sensitive	S4W (-12.31)	S4T (-0.78)
5	P	-1.96	Tolerant	P5W (-7.88)	P5A (3.43)
6	A	-5.127	Sensitive	A6W (-9.96)	A6E (-0.59)
7	D	-10.465	Critical	D7W (-14.32)	D7E (-1.05)
8	K	-5.023	Sensitive	K8W (-9.15)	K8V (-1.21)
9	T	-3.982	Tolerant	T9W (-10.53)	T9A (2.13)
10	N	-1.165	Tolerant	N10W (-6.51)	N10A (2.64)
11	V	-8.676	Critical	V11H (-12.32)	V11I (1.13)
12	K	-4.754	Sensitive	K12W (-10.57)	K12Q (-0.94)
13	A	-5.882	Sensitive	A13W (-11.48)	A13S (-1.33)
14	A	-3.053	Tolerant	A14D (-7.08)	A14I (3.37)
15	W	-11.563	Critical	W15D (-15.44)	W15F (-4.46)

Figure 1. The DeepMutate-3D interface, scanning haemoglobin subunit alpha (UniProt P69905, 142 residues) with ESM-2 650M under masked-marginal scoring. (A) The complete interface: sequence and accession inputs, model and scoring controls, CSV and PDB downloads, and a summary reporting the device, model, inference time and the most constrained residues. The scan shown ran on dynamically allocated NVIDIA hardware in 3.99 s, 28.1 ms per residue. (B) The per-residue table, giving the sensitivity score, a categorical verdict, and the most damaging and most tolerated substitution at each position. (C) The AlphaFold model painted by residue sensitivity, with the hover readout for H88. UniProt annotates this position as the proximal haem b binding residue, and the scan ranks it third most constrained of 142 positions, in the top 2%, having seen only the sequence. Red marks constrained positions and blue substitution-tolerant ones, with the

neutral midpoint fixed at zero so colour is comparable between proteins. Panels B and C are laid out as the application presents them.

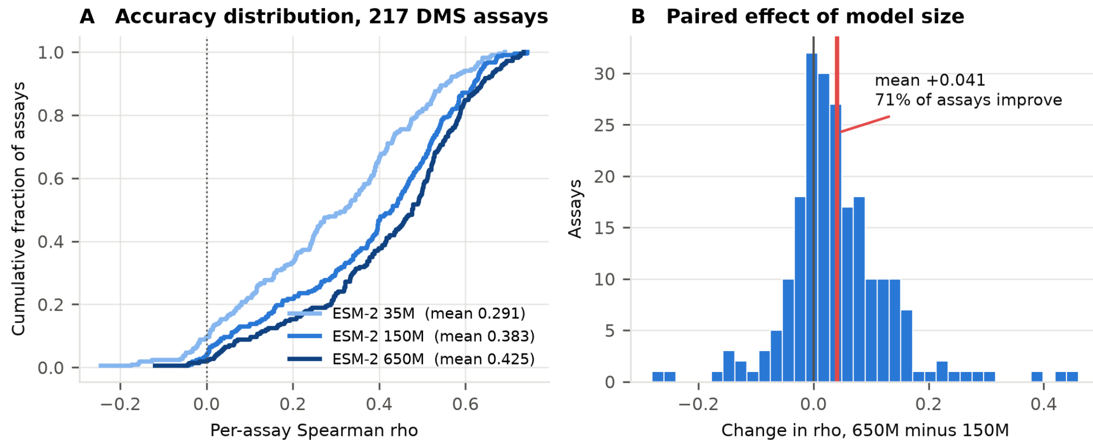


Figure 2. Effect of model size on ProteinGym. (A) Empirical cumulative distribution of per-assay Spearman correlation for three ESM-2 sizes across 217 assays. The distribution shifts uniformly rightward with model capacity rather than improving only on easy assays. (B) Paired per-assay difference between 650M and 150M. Pairing removes between-assay variance; the mean gain is +0.041 and 71% of assays improve.

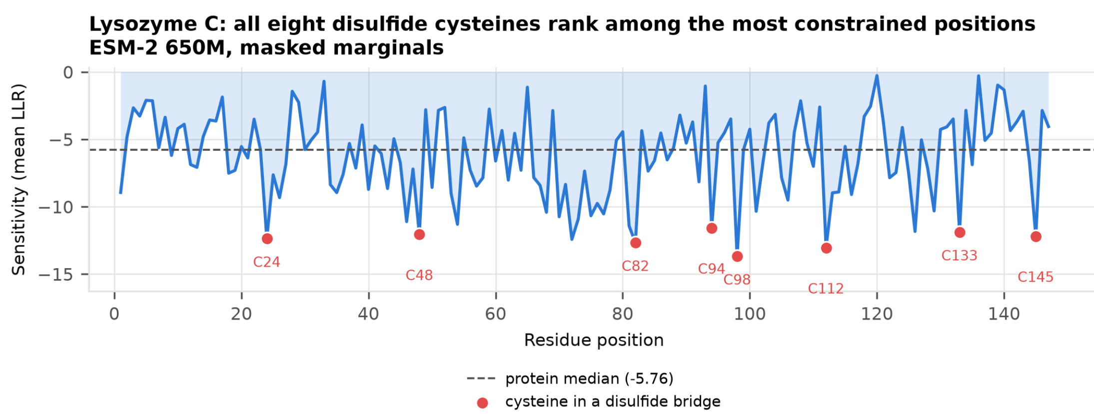


Figure 3. Sensitivity along the lysozyme C sequence. Per-residue sensitivity for all 147 positions under ESM-2 650M with masked marginals, with the eight cysteines forming the four UniProt-annotated disulfide bridges marked. They occupy ranks 1, 2, 3, 5, 6, 7, 8 and 10 of 147. Under wildtype marginals six of the eight fall in the ten most constrained positions. The dashed line is the protein median.

8. Availability

Source code, validation scripts, and all result files are at https://github.com/Saadman/DeepMutate-3D under Apache-2.0. The web application is at https://huggingface.co/spaces/ras1992/DeepMutate-3D. Every table in this note is reproducible from the scripts in `validation/` and `examples/`.

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